

Planning-Quality Degradation in Long-Horizon Autonomous Research Agents

Abstract

Autonomous agents built on large language models (LLMs) are increasingly assigned multi-step, open-ended research work — reviewing literature, designing and running experiments, and drafting manuscripts — with limited human supervision. As the length of the task horizon such agents are asked to manage grows, evidence from benchmarks and controlled studies converges on a consistent pattern: the quality of the plans an agent forms, revises, and executes deteriorates in ways distinct from, and often more consequential than, a simple rise in per-step error rate. This paper synthesizes evidence for this phenomenon, termed here planning-quality degradation, and organizes its documented causes into five interacting mechanisms: context-induced attention decay, autoregressive error compounding, myopic step-wise reasoning in place of genuine lookahead, cross-module and cross-agent failure propagation, and research-specific methodological drift unique to autonomous scientific-discovery systems. Current mitigations — explicit lookahead and value-propagation methods, hierarchical subgoal decomposition, structured context management, and reflection-based self-correction — are reviewed and shown to address only parts of the underlying problem. The paper then identifies measurement, benchmarking, and oversight challenges that constrain progress, and proposes a research agenda centered on standardized long-horizon research-workflow benchmarks, mechanistic accounts of context-length effects, and planning-quality metrics separable from raw task success. The analysis suggests that planning-quality degradation is a structural property of current autoregressive agent architectures operating over long horizons, with direct implications for the reliability of autonomous research agents specifically.

Keywords: long-horizon planning; autonomous agents; large language models; context degradation; error accumulation; agentic AI; automated scientific discovery; planning benchmarks

1. Introduction

Large language model (LLM) agents have moved rapidly from single-turn question answering toward multi-step, tool-using systems operating with limited human oversight over spans of minutes, hours, and increasingly days. Systems such as The AI Scientist (Lu et al., 2024) and Agent Laboratory (Schmidgall et al., 2025) now automate substantial portions of the research lifecycle, from literature review and hypothesis generation to experiment execution and manuscript drafting. Model Evaluation and Threat Research (METR) has quantified the underlying capability trend directly: the length of software and research tasks that frontier agents can complete with 50% reliability has doubled roughly every seven months since 2019, with the strongest models in their study handling tasks that take skilled humans on the order of one to two hours (Kwa et al., 2025). Extrapolated forward, this trend implies agents that can autonomously complete month-long projects within several years.

This trajectory of increasing task-length capability coexists with a persistent, well-documented finding: agent performance does not degrade smoothly as horizons lengthen. A specific and separable capability — the agent’s ability to form, maintain, revise, and execute a coherent plan across many interdependent steps — deteriorates in ways not fully captured by end-task accuracy alone. On GAIA, a benchmark of otherwise human-easy multi-step assistant tasks, human respondents scored 92% while GPT-4 augmented with plugins scored 15% (Mialon et al., 2024). On SWE-bench, drawn from real GitHub issues requiring coordinated multi-file reasoning, the best-performing model at benchmark introduction resolved only

1.96% of issues (Jimenez et al., 2024). On TravelPlanner, which requires jointly satisfying many simultaneous constraints across a multi-day itinerary, GPT-4 achieved a final-plan success rate of 0.6% despite far higher rates of satisfying any single constraint in isolation (Xie et al., 2024). In each case the gap between component-level competence and end-to-end planning success widens as the number of interdependent steps grows.

We refer to this decline as planning-quality degradation: a measurable reduction in the coherence, validity, adaptivity, and goal-alignment of an agent’s plan as a function of trajectory length, context growth, or horizon depth, distinguishable from a simple accumulation of independent per-step errors. The distinction matters because it points toward structural causes — how autoregressive models allocate attention over growing histories, how step-wise scoring induces short-sighted commitments, and how errors in one functional module (memory, reflection, action selection) propagate into others — rather than a single fixable defect. It is of particular consequence for autonomous research agents specifically, because a degraded plan in this setting does not merely fail a task; it can yield a scientific artifact — a benchmark comparison, a statistical claim, a written justification — that is fluent and well formatted yet methodologically unsound, as recent audits of AI-scientist systems demonstrate (Luo, Kasirzadeh, & Shah, 2025).

This paper makes three contributions. First, it synthesizes evidence for planning-quality degradation across general-purpose LLM agent benchmarks, controlled long-horizon studies, and autonomous research-agent systems. Second, it organizes the underlying mechanisms into an integrative taxonomy spanning context effects, error compounding, myopic reasoning, cross-module propagation, and research-specific methodological drift. Third, it surveys current mitigation strategies against this taxonomy and identifies where they remain partial, before outlining a research agenda targeting the most consequential gaps.

The remainder of the paper proceeds as follows. Section 2 situates the problem within the literature on LLM agent reasoning, planning benchmarks, and autonomous research systems. Section 3 develops the taxonomy of degradation mechanisms. Section 4 reviews current technical approaches. Section 5 discusses measurement and methodological challenges. Section 6 identifies research gaps and future directions, and Section 7 concludes.

2. Background, Related Work, and Literature Review

2.1 Reasoning-acting frameworks and the planning question

Contemporary LLM agents build on a small number of foundational prompting and control paradigms. Chain-of-thought prompting demonstrated that eliciting intermediate reasoning steps substantially improves performance on arithmetic, commonsense, and symbolic reasoning tasks (Wei et al., 2022). ReAct extended this idea to interactive settings by interleaving reasoning traces with executable actions, allowing an agent to update its plan in light of environment feedback and reducing — though not eliminating — hallucination and error propagation relative to reasoning alone (Yao, Zhao, et al., 2023). Tree of Thoughts generalized single-chain reasoning into a deliberate search process that maintains and evaluates multiple candidate reasoning paths, explicitly modeling problem solving as search with lookahead and backtracking (Yao, Yu, et al., 2023). Reflexion added a further layer, allowing an agent to verbally critique its own failed attempts and carry that critique forward as episodic context for subsequent trials, functioning as a lightweight substitute for weight updates (Shinn et al., 2023).

Each of these frameworks improves short-horizon task performance, but none was designed with an explicit account of how coherence should be maintained across dozens or hundreds of sequential decisions. A more

skeptical strand of work argues that autoregressive LLMs, whatever their fluency, do not perform planning or self-verification in the sense used in classical AI planning, and are better characterized as approximate knowledge sources that can support planning only when coupled with external verifiers — an architecture referred to as “LLM-Modulo” (Kambhampati et al., 2024). This critique is grounded empirically in PlanBench, a benchmark built on classical planning domains specifically designed to separate genuine plan construction from retrieval of memorized commonsense plans, on which contemporary LLMs perform poorly without external plan validators (Valmeekam et al., 2023).

2.2 Benchmarking long-horizon and multi-constraint agent behavior

A second literature stream builds benchmarks that stress multi-step, tool-using agent behavior directly rather than isolated reasoning steps. GAIA poses real-world assistant questions that are conceptually simple for humans yet require multi-step tool use, reasoning, and multi-modal handling (Mialon et al., 2024). SWE-bench operationalizes long-horizon reasoning within real software repositories, requiring models to localize, understand, and coordinate edits across multiple files based on natural-language issue descriptions (Jimenez et al., 2024). TravelPlanner isolates a distinct failure pattern: agents that satisfy individual constraints in isolation fail to jointly satisfy many constraints simultaneously across an extended plan, with failures attributed to agents losing focus on the overall task, misusing information-gathering tools, and losing track of previously satisfied or violated constraints as constraint count grows (Xie et al., 2024). METR’s time-horizon methodology reframes the underlying capability trend in units directly comparable to human labor, showing exponential growth in the length of tasks agents can complete autonomously, while explicitly noting that this metric captures overall task-length capability rather than the internal quality of the plans that produce success or failure (Kwa et al., 2025).

2.3 Autonomous research agents

A distinct and rapidly growing literature applies agentic LLM systems to the scientific research process itself. The AI Scientist automates ideation, experimental implementation, and manuscript composition end to end, including a simulated peer-review step (Lu et al., 2024). Agent Laboratory structures the workflow into literature review, experimentation, and report-writing phases carried out by specialized agent roles working alongside a human collaborator (Schmidgall et al., 2025). Broader syntheses of LLM-based autonomous agents (Wang et al., 2024) and of LLM agent planning specifically (Huang et al., 2024) both note that task decomposition, memory management, and reflection are recurring design levers, but that none fully resolves degradation over long horizons. Memory-augmented architectures such as MemGPT, which manages a bounded context window analogously to virtual memory in an operating system (Packer et al., 2023), and Voyager, which builds an ever-growing, reusable skill library to support lifelong learning in an open-ended environment (Wang, Xie, et al., 2023), represent early attempts to decouple agent competence from raw context-window size. Auditing work specific to autonomous research agents, discussed further in Section 3.5, shows that automating the research workflow introduces methodological failure modes that are largely orthogonal to conventional notions of task success (Luo, Kasirzadeh, & Shah, 2025).

Taken together, this literature establishes two things relevant to the present paper: first, that longer horizons reliably expose an agent capability gap that shorter-horizon benchmarks do not; and second, that this gap has been characterized piecemeal — as a context problem, a reasoning problem, a memory problem, or a research-methodology problem — without a unifying account of how these characterizations relate. Section 3 develops such an account.

3. Main Discussion: A Taxonomy of Planning-Quality Degradation Mechanisms

We organize the evidence into five mechanisms, presented roughly in the order in which they compound within a single long-horizon trajectory: context-induced degradation, error compounding, myopic reasoning in place of lookahead planning, cross-module and cross-agent failure propagation, and research-specific methodological drift. These mechanisms are not mutually exclusive; a single failed trajectory typically reflects several acting together.

3.1 Context-induced degradation

The most extensively documented mechanism concerns how model performance changes as the input context an agent must condition on grows, independent of any change in task difficulty. Liu et al. (2024) showed that language models exhibit a U-shaped performance curve on multi-document question answering: relevant information positioned at the beginning or end of a long context is used reliably, while the same information placed in the middle is frequently missed, indicating that models do not access long contexts uniformly. A large-scale evaluation across eighteen frontier models — including GPT-4.1, the Claude 4 family, Gemini 2.5, and Qwen3 — found that accuracy degrades non-uniformly as input length grows, sometimes by 30 to 50 percent well before a model’s documented context limit is reached, and that this degradation is driven more by the semantic similarity between relevant and irrelevant content than by length alone (Hong, Troynikov, & Huber, 2025). For long-horizon agents this effect compounds directly: every tool call, file read, and retrieved document accumulates in the transcript, so the agent’s effective reasoning budget over what matters shrinks turn by turn even when the nominal context window is far from full. The consequences extend beyond task success: in a safety-oriented study, LLM-based monitors tasked with detecting a subtly dangerous action embedded in an otherwise benign coding-agent transcript missed the action two to thirty times more often when it occurred after 800,000 tokens of prior benign activity than when evaluated in isolation, showing that context-induced degradation also erodes an agent’s or overseer’s ability to notice its own planning failures at scale (Martin & Roger, 2026).

3.2 Autoregressive error compounding

A second, closely related mechanism concerns how small errors, once introduced, propagate and amplify across subsequent generation steps rather than remaining isolated. Because autoregressive models condition each new output on their own prior outputs, an early mistake perturbs the distribution of states the agent subsequently reasons over, and later decisions are made against this already-shifted distribution — a dynamic shown empirically to scale poorly with trajectory length in language-model rollouts (Arbuzov, Bei, Dong, Kalaev, & Shvets, 2025). A specific instance of this dynamic is hallucination snowballing: Zhang, Press, Merrill, Liu, and Smith (2023) showed that ChatGPT and GPT-4, when justifying an initial incorrect answer, frequently generate additional false claims that they can separately recognize as false in isolation (identified correctly 67% and 87% of the time, respectively, when asked afterward), indicating that the model over-commits to an early mistake rather than lacking the relevant knowledge. In agentic settings this compounding is structurally worse because agents are non-Markovian systems whose decisions depend on the full accumulated history of context, retrieved evidence, and prior actions; flawed intermediate artifacts are recycled into later decision contexts, so systematic planning error can emerge even when each individual step appears locally reasonable in isolation.

3.3 Myopic step-wise reasoning versus genuine lookahead

A third mechanism is more specifically about planning as distinct from reasoning or memory. Controlled analysis of LLM agents in deterministic, fully observable planning environments found that step-wise reasoning policies — including chain-of-thought and beam-search variants — behave as a form of greedy, locally scored policy: they favor actions that look attractive at the current step without adequately weighting

delayed consequences, commit to a suboptimal trajectory prefix within the first few decisions, and rarely recover thereafter, with wider beam search delaying but not preventing this early, largely irreversible commitment (Wang, Wu, et al., 2026). This finding gives a mechanistic account of why apparently strong step-by-step reasoning capability, as elicited by chain-of-thought prompting (Wei et al., 2022), does not translate into coherent long-horizon planning: reasoning improves the quality of any single inference, but genuine planning additionally requires future-aware evaluation — propagating information about downstream consequences back to influence earlier choices — which step-wise scoring does not provide. This is consistent with the broader claim that LLMs alone do not perform planning or self-verification and instead require an explicit lookahead or external verification loop to do so reliably (Kambhampati et al., 2024). A related but contested body of evidence extends this concern to explicit “reasoning” models: Shojaei et al. (2025) reported that large reasoning models exhibit an abrupt accuracy collapse on controllable planning puzzles beyond a certain complexity threshold, with reasoning effort itself declining near the collapse point. A subsequent comment argued that this pattern reflects experimental design limitations — models running into output-token limits and being unable to enumerate long solutions rather than failing to reason about them — rather than a fundamental planning failure (Lawsen, 2025), illustrating that even the empirical characterization of degraded long-horizon planning quality remains an active methodological dispute (discussed further in Section 5).

3.4 Cross-module and cross-agent failure propagation

A fourth mechanism operates at the level of agent architecture rather than any single reasoning step. Where agents are decomposed into functional modules — memory, reflection, planning, and action — or into multiple communicating agents, a root-cause error in one module or agent readily cascades into others. A large-scale annotation study of over 1,600 traces across seven popular multi-agent frameworks found overall failure rates between 41% and 86.7%, and derived a taxonomy of fourteen distinct failure modes clustered into system design issues, inter-agent misalignment, and inadequate task verification (Cemri et al., 2025). A complementary single-agent study decomposing hundreds of trajectories from ALFWorld, GAIA, and WebShop into memory, reflection, planning, and action modules identified error propagation as the central bottleneck to agent reliability, showing that a single early root-cause failure typically cascades into a sequence of compounding, increasingly severe downstream errors that are progressively harder to attribute back to their origin (Zhu et al., 2025). Reflection mechanisms intended to counteract this — having an agent verbally critique and revise its own trajectory (Shinn et al., 2023) — can, without safeguards, themselves become a channel for propagation: because reflective text is folded back into the agent’s own context, an early biased or mistaken self-critique can stabilize and reinforce rather than correct the original error across iterations.

3.5 Research-specific methodological drift

A final mechanism is specific to agents whose task is scientific research rather than software or web navigation, and concerns the quality of the plan not merely as a sequence of actions but as an evidentiary and methodological argument. Systematic audits of two prominent open-source AI-scientist systems identified four recurring failure modes that degrade the integrity of automatically produced research even when the agent’s stated experimental steps proceed without an overt crash: inappropriate benchmark selection that inflates reported performance through favorable dataset choice, data leakage between training and evaluation splits, misuse or misapplication of evaluation metrics, and post-hoc selection bias in which positive results are selectively reported in a manner functionally equivalent to fitting to the test set (Luo, Kasirzadeh, & Shah, 2025). These are planning-quality failures in the sense developed here: the agent forms and executes a plan that is locally coherent and produces a fluent, well-formatted manuscript, yet the plan

is invalid as a piece of scientific reasoning because it does not track the evidentiary constraints a human researcher would apply at each step. Because the artifacts produced — tables, statistics, prose justification — look identical in surface form whether or not these constraints were respected, this mechanism is comparatively harder to detect than an outright task failure, and it compounds with the other four mechanisms across a full research pipeline that may span literature review, coding, experiment execution, and writing.

4. Current Approaches and Developments

Mitigation efforts map fairly directly onto the taxonomy above, though most target a single mechanism rather than the interaction among them.

Against myopic step-wise reasoning, the most direct intervention is to introduce explicit future-aware evaluation. FLARE (Future-aware Lookahead with Reward Estimation) augments a step-wise policy with a search tree, value propagation, and limited commitment, allowing downstream simulated outcomes to influence earlier decisions; across multiple benchmarks and backbones this allowed a substantially smaller open model to outperform a much larger model using standard step-by-step reasoning, and delayed and reduced — though did not eliminate — early, hard-to-recover commitments (Wang, Wu, et al., 2026). Tree of Thoughts similarly reintroduces deliberate search and backtracking over reasoning paths in place of a single committed chain (Yao, Yu, et al., 2023), and the LLM-Modulo framework proposes pairing LLMs with external, sound plan verifiers rather than relying on the LLM to self-verify, treating the model as a candidate-plan generator rather than an authoritative planner (Kambhampati et al., 2024).

Against context-induced degradation, current work pursues structured management of what enters an agent’s working context rather than simply extending window size. Hierarchical approaches decompose a long-horizon task into subgoals and periodically fold or summarize the execution history of completed subgoals out of active context, targeting the state-tracking failure that arises when continuously growing histories dilute an agent’s awareness of its current task stage. Related subgoal-driven frameworks apply online subgoal decomposition together with reinforcement learning using milestone-based dense rewards to keep long-horizon web agents oriented toward a final goal across extended interaction sequences. More broadly, memory-augmented architectures such as MemGPT manage a bounded context analogously to virtual memory, paging information in and out as needed (Packer et al., 2023), while Voyager maintains a persistent, ever-growing skill library external to the context window itself so that competence compounds without requiring the full interaction history to remain in context (Wang, Xie, et al., 2023).

Against error compounding and hallucination snowballing, reflection-based self-correction remains the dominant approach: Reflexion has an agent verbally critique failed attempts and retain that critique as episodic memory for subsequent trials, improving performance without weight updates (Shinn et al., 2023). This mitigation is partial, however, since self-conditioning on one’s own prior errors can, in the absence of independent verification, reinforce rather than correct systematic mistakes — precisely the dynamic underlying hallucination snowballing (Zhang et al., 2023).

Against cross-module and cross-agent propagation, the principal current development is diagnostic rather than corrective: taxonomies such as MAST (Cemri et al., 2025) and AgentErrorTaxonomy together with the accompanying AgentErrorBench dataset (Zhu et al., 2025) enable systematic, root-cause attribution of failures across memory, reflection, planning, and action modules, and support debugging tools that trace a downstream failure back to its originating module rather than treating failure as an undifferentiated

outcome. This diagnostic capability is a necessary precursor to targeted correction but does not by itself prevent propagation.

Against research-specific methodological drift, mitigation is the least developed of the five areas. The primary current approach is auditing after the fact — applying controlled detection tasks for benchmark-selection bias, data leakage, metric misuse, and post-hoc selection bias to the outputs of AI-scientist systems (Luo, Kasirzadeh, & Shah, 2025) — together with retaining a human collaborator in the loop at key checkpoints, as in Agent Laboratory’s phase structure (Schmidgall et al., 2025). Neither constitutes a built-in safeguard within the agent’s own planning loop.

5. Research Challenges and Limitations

Several challenges limit how confidently the field can currently characterize, compare, and correct planning-quality degradation.

First, measurement and attribution are confounded. Horizon length, task complexity, number of available tools, and context length tend to co-vary in real benchmarks, making it difficult to isolate which factor is responsible for an observed decline. The PlanBench-based controlled study of step-wise versus lookahead policies addresses this by using deterministic, fully observable environments with known optimal solutions (Wang, Wu, et al., 2026), but such controlled settings necessarily sacrifice the ecological realism of GAIA-, SWE-bench-, or TravelPlanner-style tasks, so findings from one setting do not automatically transfer to the other.

Second, even within a single well-controlled study, interpretation is contested. The dispute between Shojaei et al. (2025), who reported an abrupt planning-accuracy collapse in reasoning models beyond a complexity threshold, and Lawsen (2025), who argued the same data reflects output-length and evaluation-harness artifacts rather than a reasoning failure, illustrates that the field currently lacks agreed-upon protocols for distinguishing genuine planning-quality degradation from artifacts of how a benchmark scores incomplete or truncated outputs. This is a general problem for the area: many long-horizon benchmarks score binary task success, which conflates “the agent could not plan this” with “the agent’s correct plan was cut off, mis-parsed, or scored incorrectly.”

Third, standardized benchmarks for the specific object of interest in this paper — autonomous research-agent planning quality, as distinct from software-engineering or web-navigation planning quality — remain immature. Most quantitative evidence on autonomous research agents currently comes from case-study audits of a small number of open-source systems (Luo, Kasirzadeh, & Shah, 2025) rather than from standardized benchmarks with graded task lengths analogous to GAIA or SWE-bench, limiting the ability to compare systems, track progress over time, or attribute observed methodological drift to specific architectural choices.

Fourth, evaluating long-horizon behavior is expensive and rapidly non-stationary. METR’s own time-horizon methodology required timing human experts on curated tasks and evaluating multiple frontier models across many trials to obtain statistically defensible estimates (Kwa et al., 2025), a cost structure that scales poorly as horizons of interest extend from hours toward days or weeks. Compounding this, the underlying models are updated on a cadence of months, so longitudinal comparisons risk becoming stale before they can be fully analyzed and replicated.

Fifth, planning-quality degradation has direct implications for oversight, not only for task success. As transcripts lengthen, the same context-induced degradation that erodes an agent’s own planning also erodes

the ability of LLM-based monitors to detect subtly dangerous or erroneous actions embedded within an otherwise benign long transcript (Martin & Roger, 2026), meaning that the tools most readily available for catching degraded plans in practice are themselves subject to the phenomenon they are meant to detect.

6. Research Gaps and Future Directions

Building on the challenges above, we identify five priority directions.

First, the field needs standardized, graded long-horizon benchmarks specific to research workflows — spanning literature synthesis, experiment design, execution, and write-up — analogous in rigor to GAIA (Mialon et al., 2024) and SWE-bench (Jimenez et al., 2024) but constructed so that task length, tool count, and context length can be varied independently, enabling controlled attribution of degradation to specific structural causes rather than to horizon length as an undifferentiated variable.

Second, mechanistic accounts of context-length effects on attention allocation remain underdeveloped relative to their practical importance. Current evidence establishes the existence and rough shape of context-induced degradation, including the U-shaped positional effect (Liu et al., 2024) and its non-uniform, similarity-dependent form across contemporary frontier models (Hong et al., 2025), but the interpretability work needed to predict when and why a given architecture will exhibit this pattern, and to design targeted fixes rather than post hoc context management, remains at an early stage.

Third, the field lacks planning-quality metrics that are separable from raw task success. Because binary success rates conflate genuine planning failure with scoring or truncation artifacts (Section 5), metrics that directly measure plan coherence, revision cost, constraint-satisfaction rate under increasing constraint count, and recovery rate after an induced early error — in the spirit of the position-of-first-error and post-error-recoverability measures used in controlled lookahead studies (Wang, Wu, et al., 2026) — would allow more precise comparison across mitigation strategies than end-task accuracy alone.

Fourth, integrating symbolic or externally verifiable planning components with LLM-generated candidate plans, as proposed under the LLM-Modulo framing (Kambhampati et al., 2024), remains comparatively unexplored specifically within autonomous research-agent pipelines, where the analogue of a sound plan verifier might include automated checks for data leakage, pre-registration-style commitment to evaluation metrics before results are seen, and statistical-validity checking — directly targeting the methodological-drift mechanism identified in Section 3.5 rather than only the step-sequencing mechanisms addressed by most current planning research.

Fifth, persistent, cross-session epistemic state for research agents is an open architectural problem. Current context-management and memory approaches (Packer et al., 2023; Wang, Xie, et al., 2023) primarily address within-session degradation, but autonomous research characteristically spans multiple sessions and requires an evolving, explicit, and auditable representation of what has been tried, what has been confirmed or refuted, and why — closer to a persistent lab notebook than a rolling context window — for which current single-session context-management techniques were not designed.

7. Conclusion

Across general-purpose agent benchmarks, controlled planning studies, and autonomous research-agent audits, the evidence converges on a common pattern: as the horizon over which an LLM-based agent must plan lengthens, plan quality degrades through several interacting mechanisms rather than through a single fixable defect. Context-induced attention decay, autoregressive error compounding, myopic step-wise

reasoning in place of genuine lookahead, cross-module and cross-agent failure propagation, and — specifically for research agents — methodological drift in the scientific reasoning embedded in a plan, together account for a substantial share of the gap between short-horizon competence and long-horizon reliability documented throughout this paper. Current mitigations, from explicit lookahead search and hierarchical subgoal decomposition to structured context management and reflection-based self-correction, each address one or two of these mechanisms and have demonstrated measurable gains, but none resolves the phenomenon as a whole, and the field's own methodological tools for measuring degradation remain contested, as the dispute over reasoning-model accuracy collapse illustrates. For autonomous research agents in particular, the stakes of unresolved planning-quality degradation extend beyond task failure: a degraded plan can surface as a fluent but methodologically unsound scientific artifact that is difficult to distinguish from a sound one without independent audit. Closing this gap will likely require coordinated progress on standardized long-horizon research benchmarks, mechanistic understanding of context effects, planning-quality metrics separable from task success, and externally verifiable planning architectures purpose-built for the evidentiary demands of scientific reasoning rather than adapted from software- or web-agent settings.

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